Report

Assignment 4

Data Analytics E0-259

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Implementation Summary

The objective of this assignment is to predict the required runs when overs are lost due to unforeseen circumstances (e.g., rain delays) using a modified Duckworth-Lewis (DLS) method. The model predicts runs based on the remaining resources, which are determined by the number of overs left and wickets in hand and minimize error between predicted and actual runs.

Approach:

1. **Run Production Function/ Model Function:** The run production function is modelled as:
   - $Z(u, w) = Z_0(w)[1 - \exp\{-L(w)u/Z_0(w)\}]$. Z0(w): Asymptotic value of runs with w wickets in hand.
   - L(w): Rate parameter for each wicket.
   - u: Overs remaining.
2. **Loss Function:** The loss function, which measures the difference between predicted runs y′ and actual runs y, is defined as:

   $Loss(y', y) = (y'+1)\log((y'+1)/(y+1)) - y' + y,$
   This is used to compute the error across overs and wickets in the first innings.

3. **Optimization Strategy:**
   - Initial guess for Z0(w) was obtained through averaging the maximum scores of every match's first innings at each wicket.
   - A nonlinear optimization technique (L-BFGS-B) was used to minimize the loss function for each Z0(w) and L(w) as it handles box constraints like ensuring non negative parameters and is memory efficient.
   - After obtaining preliminary values for L(w), a common L was computed by taking the weighted average of all L(w) values, weighted by the number of data points for each wicket.
4. **Final Optimization:** Once the common slope L was calculated, the final model was fitted by keeping L fixed and optimizing only for Z0(w) values using the same optimization method.

Data Handling:

1. **Data Loading and Preprocessing:**
   - The dataset containing first innings data from cricket matches was loaded.
   - Preprocessing steps involved cleaning the data, handling missing or inconsistent entries, and clipping out-of-bound values.

- The data points were split into overs (runs.remaining) and wickets in hand, with runs scored as the target variable.

2. **Clipping Values -** To avoid issues such as taking the logarithm of zero, the true runs were clipped to avoid numerical errors.

Assumptions:

- Initial values for Z0(w) were assumed to be the average of the maximum first-innings scores for each wicket across all matches.
- The common L was computed based on the assumption that the slopes across different wickets converge to a single value.

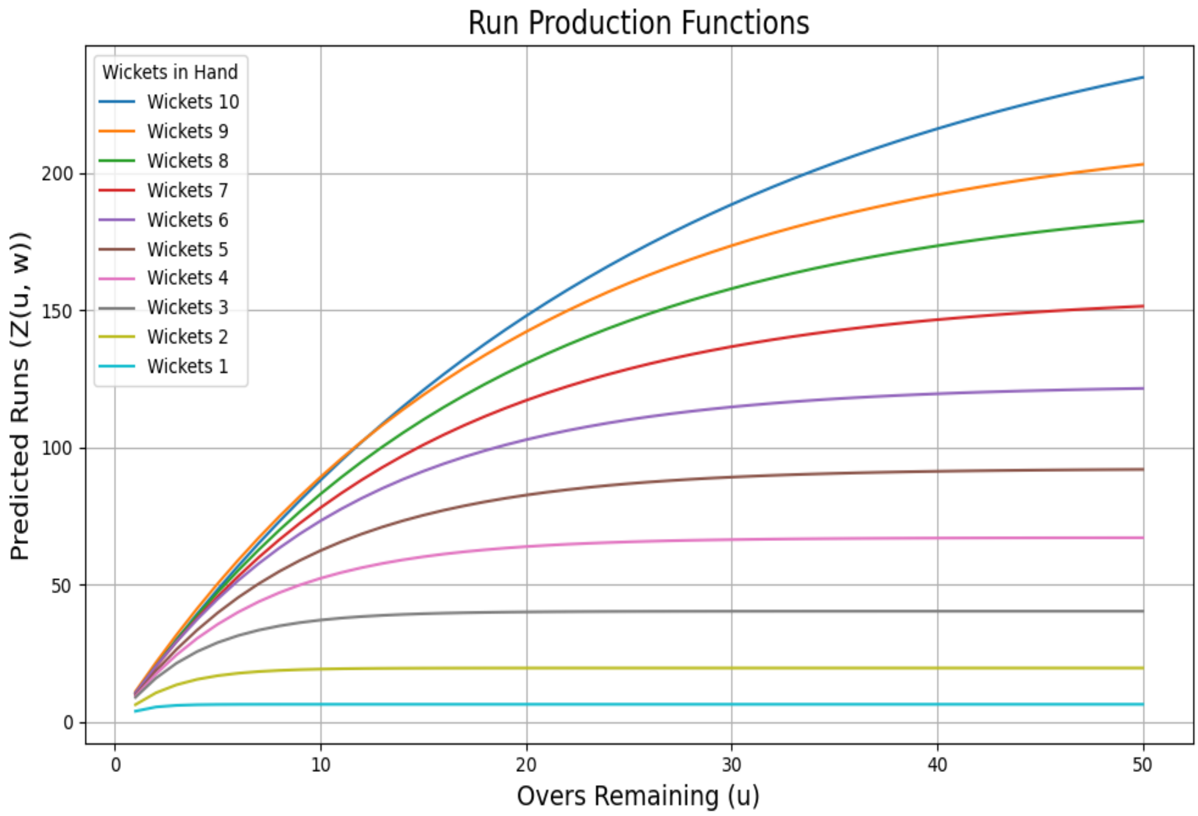
Implementation Steps:

1. **Initial Parameter Estimation:**
    - Preliminary Z0(w) and L(w) were estimated for each wicket using nonlinear optimization, minimizing the loss function.
2. **L Calculation:**
    - A common L was calculated as a weighted average of the preliminary L(w) values, with the number of data points for each wicket serving as the weights.
3. **Final Parameter Estimation:**
    - With the common L fixed, final Z0(w) values were estimated through optimization.
4. **Normalized Loss Calculation:**
    - The total loss was normalized by dividing the sum of the losses across all data points by the total number of data points (overs and wickets). This provided a measure of the model's overall performance.

Output:

Preliminary Parameters

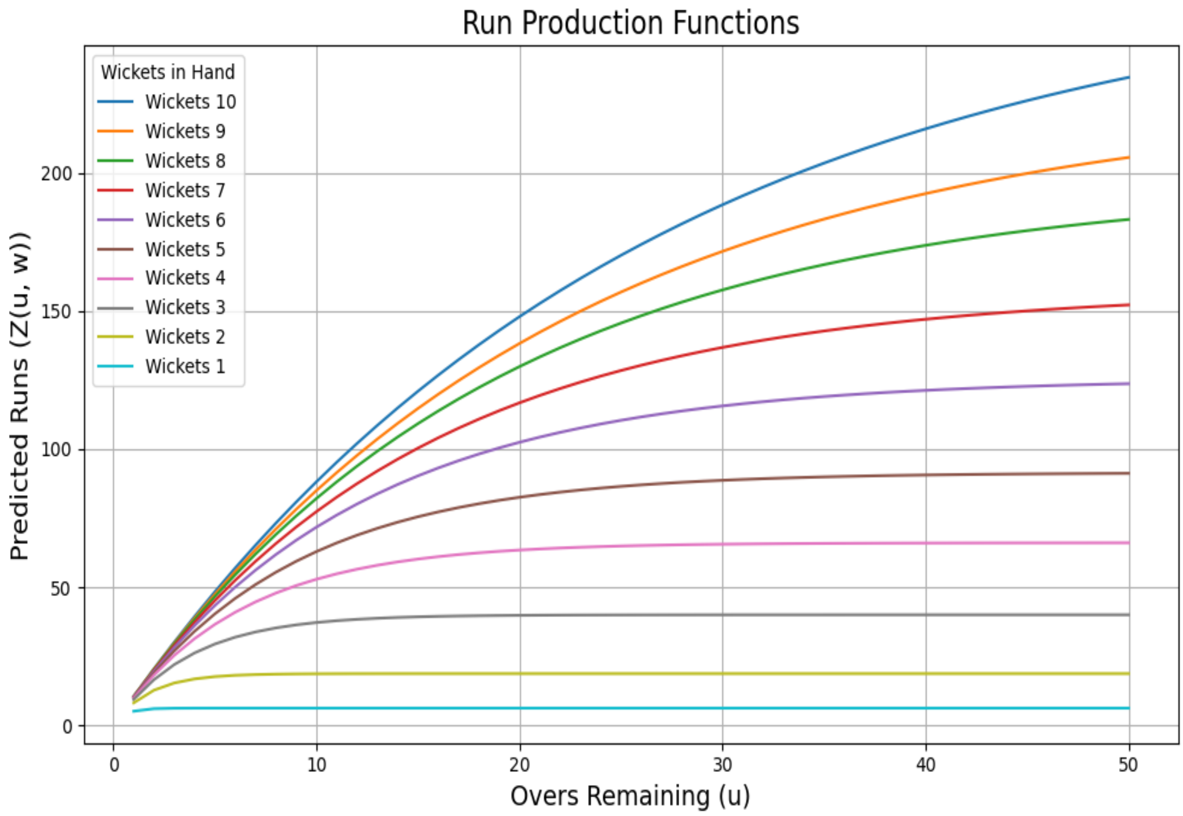
Wicket	Z0_w	L_w
10	274.19	10.62
9	219.05	11.45
8	194.27	10.83
7	165.31	10.81
6	122.77	11.15
5	92.31	10.41
4	67.15	10.16
3	40.38	10.21
2	19.71	7.70
1	6.50	6.17



Final Parameters

Wicket	Z0_w
10	273.85
9	227.60
8	196.21
7	157.61
6	125.50
5	91.51
4	66.10
3	40.02
2	18.72
1	6.22

Common L is 10.64



The obtained normalised loss is 6.261436857220674

Conclusion

The implementation successfully predicted run production using a modified DLS method, with parameters optimized for each wicket and overs remaining. The model's predicted runs deviate from the actual runs by an average of **6.26 units** (runs) for each over and wicket combination.